WORKED SOLUTION

STEP I 1989, Question 5

Pure • Binomial Coefficient Identities

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Read the question before you go on. It is in the STEP Support Programme's question database:

step.maths.org/questions/89-s1-q5

The key idea

The binomial expansion is an identity in x, so it stays true whatever number we put in. Choosing $x = 1$ adds up all the coefficients, and $x = -1$ adds them with alternating signs; combining the two separates the even-numbered from the odd-numbered ones. For the weighted sum in (ii), the factor r in front of $\binom{n}{r}$ is exactly what differentiating x^r produces, so we differentiate the expansion and then substitute.

Solution

For a positive integer n,

$$(1+x)^n = \sum_{r=0}^{n} \binom{n}{r} x^r = \binom{n}{0} + \binom{n}{1}x + \binom{n}{2}x^2 + \cdots + \binom{n}{n}x^n,$$

where $\binom{n}{r} = \dfrac{n!}{r!\,(n-r)!}$. This holds for every real x.

Part (i)

Write $E = \binom{n}{0} + \binom{n}{2} + \cdots + \binom{n}{n}$ for the sum of the even-numbered coefficients and $O = \binom{n}{1} + \binom{n}{3} + \cdots + \binom{n}{n-1}$ for the odd-numbered ones. Because n is even, the last term $\binom{n}{n}$ belongs to E and $\binom{n}{n-1}$ to O, so between them E and O contain every coefficient exactly once.

Putting $x = 1$ in the expansion gives

$$E + O = 2^n.$$

Putting $x = -1$, each even power $(-1)^r$ is $+1$ and each odd power is -1, while the left-hand side is $0^n = 0$ since $n \geq 1$. So

$$E - O = 0.$$

Hence $E = O$, and then $2E = 2^n$, so

$$E = O = 2^{n-1},$$

which is the required result.

Part (ii)

Differentiate both sides of the expansion with respect to x. The constant term vanishes and x^r becomes rx^{r-1}:

$$n(1+x)^{n-1} = \binom{n}{1} + 2\binom{n}{2}x + 3\binom{n}{3}x^2 + \cdots + n\binom{n}{n}x^{n-1}.$$

Now put $x = 1$:

$$n2^{n-1} = \binom{n}{1} + 2\binom{n}{2} + 3\binom{n}{3} + \cdots + n\binom{n}{n},$$

as required. This works for every positive integer n, odd or even.

Ask yourself Is there a way to see part (ii) without calculus?

Yes. Since $r\binom{n}{r} = r \cdot \dfrac{n!}{r!\,(n-r)!} = n \cdot \dfrac{(n-1)!}{(r-1)!\,(n-r)!} = n\binom{n-1}{r-1}$ for $1 \le r \le n$, the sum is $n\sum_{r=1}^{n}\binom{n-1}{r-1} = n \cdot 2^{n-1}$, by putting $x = 1$ in the expansion of $(1+x)^{n-1}$. In words: choosing a committee of any size from n people and then a chair from the committee is the same as choosing the chair first (n ways) and then any subset of the other $n-1$ people (2^{n-1} ways).

The final sum

Split the sum into two:

$$\sum_{r=0}^{n}(r + (-1)^r)\binom{n}{r} = \sum_{r=0}^{n} r\binom{n}{r} + \sum_{r=0}^{n}(-1)^r\binom{n}{r}.$$

The $r = 0$ term of the first sum is zero, so the first sum is exactly the sum in (ii), namely $n2^{n-1}$. The second sum is the expansion of $(1+x)^n$ at $x = -1$, namely $(1-1)^n = 0$ for every positive integer n. (This is the step $E - O = 0$ from (i); it needs only $n \ge 1$, not n even.) Therefore

$$\boxed{\sum_{r=0}^{n}(r + (-1)^r)\binom{n}{r} = n2^{n-1}.}$$

Checks

- For $n = 2$: the coefficients are $1, 2, 1$, so $E = 1 + 1 = 2 = O$ and $2^1 = 2$. The sum in (ii) is $2 + 2 \cdot 1 = 4 = 2 \cdot 2^1$.

- For $n = 3$ the final sum is $(0+1)\cdot 1 + (1-1)\cdot 3 + (2+1)\cdot 3 + (3-1)\cdot 1 = 1 + 0 + 9 + 2 = 12 = 3 \cdot 2^2$.
- We checked all three identities exactly for every n from 1 to 29; for instance, $n = 6$ gives $192 = 6 \cdot 2^5$.

Common slips

- Forgetting that $\sum(-1)^r \binom{n}{r} = 0$ holds for odd n too, and so thinking the final answer depends on the parity of n. Only at $n = 0$ does it fail, and n is positive here.
- Substituting $x = 1$ before differentiating in (ii). Substitution must come last: once x is a number, there is nothing left to differentiate.

Beyond STEP

Treating a polynomial as a store of coefficients, and then substituting, differentiating or multiplying to extract information, is the idea of a *generating function*, one of the most powerful tools in combinatorics and probability. Substituting complex roots of unity instead of ± 1 picks out every third or every fourth coefficient in the same way, and differentiating a probability generating function gives the mean of a distribution, just as it gave $\sum r\binom{n}{r}$ here. Grimmett and Welsh, *Probability: An Introduction*, shows generating functions at work.

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